

taking an incoming electrical signal of a certain frequency and producing an output signal that was a multiple of that frequency.

Although these graphene chips open up a range of new applications, their practical use is limited by a very small voltage gain (typically, the amplitude of the output signal is about 40 times less than of the input signal). Moreover, none of these circuits was still demonstrated to operate at frequencies higher than 25 kHz.

Recently, researchers have been able to create graphene transistors with an on-off rate of 100 gigahertz – far exceeding the rates of previous attempts, and exceeding the speed of silicon. The 240nm graphene transistors were made using extant silicon-manufacturing equipment, meaning that for the first time graphene transistors are a conceivable – though still fanciful – replacement for silicon.

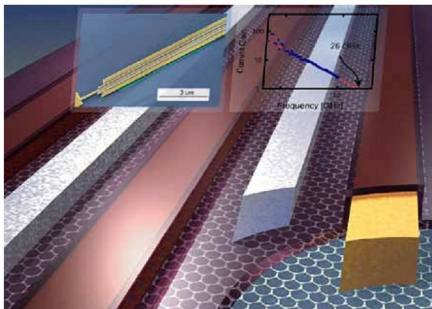
Integrated circuits. Graphene has the ideal properties to become an excellent component of integrated circuits. Its high carrier mobility and low noise allow it to be used as a channel in FETs.

However, single sheets of graphene are hard to produce, and even harder to make on top of an appropriate substrate. Researchers are looking into methods of transferring single graphene sheets from their source of origin onto a target substrate of interest.

The smallest transistor so far – one atom thick and ten-atoms wide – was made of graphene. Researchers have also fabricated and characterised graphene transistors operating at GHz frequencies. They have also created an n-type transistor, which means both n- and p-type transistors have now been created with graphene.

Besides, the researchers have demonstrated the first functional graphene integrated circuit – a complementary inverter consisting of one p- and one n-type graphene transistor. However, this inverter suffers from a very low voltage gain.

Transparent conducting electrodes.



Graphene transistor

Graphene's high electrical conductivity and high optical transparency make it a candidate for transparent conducting electrodes, required for such applications as touchscreens, liquid crystal displays, organic photovoltaic cells and organic light-emitting diodes. In particular, graphene's mechanical strength and flexibility are advantageous compared to indium tin oxide, which is brittle, and graphene films may be deposited from a solution over large areas.

Large-area, continuous, transparent and highly conducting few-layered graphene films were produced by chemical vapour deposition and used as anodes for application in photovoltaic devices. One layer of graphene absorbs 2.3 per cent of white light and this property was used to define the 'conductivity of transparency' that combines the sheet resistance and the transparency. This parameter was used to compare different materials without the use of two independent parameters.

If graphene sheets are made thin enough, these basically become transparent. This has led some experts to suggest that the material could be used in a new generation of low-cost, crack-resistant display screens for televisions and laptops. A working touch-screen display using graphene has already

been created.

Ultracapacitors. Due to the incredibly high surface-area-to-mass ratio of graphene, its one potential application is in the conductive plates of ultracapacitors. Graphene could be used to produce ultracapacitors with a greater energy storage density than is currently available.

Graphene biodevices. Graphene's modifiable chemistry, large surface area, atomic thickness and molecularly-gatable structure make antibody-functionalised graphene sheets excellent candidates for mammalian and microbial detection and diagnosis devices. The most ambitious biological application of graphene is for rapid, inexpensive electronic DNA sequencing. Integration of graphene (thickness of 0.34 nm) layers as nanoelectrodes into a nanopore can solve one of the bottlenecks of nanopore-based single-molecule DNA sequencing.

Anti-bacterial. Scientists have found that sheets of graphene oxide are highly effective at killing such bacteria as *Escherichia coli*. This means graphene could be useful in hygiene products or packaging to help keep food fresh for a long time.

Strength applications. One atom-thick sheet is not that tough, but when